Environmental Science and Pollution Research (2019) 26:31301–31314
https://doi.org/10.1007/s11356-019-06214-7

RESEARCH ARTICLE

Optimisation, experimental validation and thermodynamic study of the sequential oil extraction and biodiesel production processes from seeds of *Sterculia foetida*

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Received: 3 April 2019 / Accepted: 14 August 2019 / Published online: 30 August 2019
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Abstract

Non-edible seeds are not used in any commercial applications, which implies that they can be used for biofuel applications. The present study aimed to maximise the process conditions for oil extraction and sterculia biodiesel production from *Sterculia foetida* (*poon oil*). GC-MS identified the methyl esters of sterculia oil as sterculic acid (32%), palmitic acid (15.88%), oleic acid (10.00%), linoleic acid (9.95%) and malvalic acid (9%). Response surface methodology (RSM) based parametric optimisation of oil extraction was carried out by choosing process variables such as sample weight, volume of solvent to seed ratio and time. The optimum sample amount of 7.5 g and the volume of solvent to seed ratio of 40 mL/g resulted in a maximum oil yield of 45.27% at 3 h. The results were statistically significant ($P < 0.05$) with a regression coefficient (R^2) of 0.9988. Furthermore, the artificial neural network (ANN) resulted in an R^2 value greater than 0.9, which validates the RSM. Conventional optimisation of the temperature (55 °C), feedstock to methanol ratio (1:12), catalyst proportion (1.5%) and transesterification reaction time (60 min) yield 90.87% biodiesel production. The physicochemical characteristics of oil and biodiesel complied with the requirements of the ASTM standards. The rate constant and thermodynamic variables at the optimum temperature (333 K) were calculated from the experimental data. The activation energy (E_a), activation enthalpy in transition state theory (ΔH^{++}), activation entropy in transition state theory (ΔS^{++}) and Gibbs free energy in transition state theory (ΔG^{++}) were 37.91 kJ mol^{-1}, 35.14 kJ mol^{-1}, −239.58 J mol^{-1} K^{-1} and 79.81 kJ mol^{-1} respectively.

Keywords Poon oil · Response surface methodology · Solvent extraction · Artificial neural network · Biodiesel optimisation · Kinetics · Activation energy · Enthalpy change

Nomenclature

k	Reaction rate constant (min^{-1})
A	Arrhenius constant
E_a	Activation energy (kJ mol^{-1})
R	Universal gas constant (J K^{-1} mol^{-1})
T	Temperature in Kelvin (K)
N	Avogadro's constant (mol^{-1})
h	Planck's constant (J s)
ΔS	Entropy change (mol^{-1} K^{-1})
ΔH	Enthalpy change (J)
ΔG	Gibb's free energy (kJ mol^{-1})
ΔS^{++}	Activation entropy in transition state theory (kJ mol^{-1})
ΔH^{++}	Activation enthalpy in transition state theory (kJ mol^{-1})
ΔG^{++}	Gibbs free energy in transition state theory (kJ mol^{-1})
Yi	Equilibrium constant

Responsible editor: Ta Yeong Wu

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Introduction

The necessity for fossil fuels is increasing over time, but the availability of fossil fuels is limited, thus creating an energy crisis in the current scenario. The majority of countries are dependent on fossil fuels to fulfil their energy demands due to industrialisation. Dependency on fossil fuels leads to an adverse impact

on the economy with variation in fossil fuel prices (Dincer 2008; Kavitha and Murugavelh 2019a; Sandhya et al. 2016). The use of fossil fuels also leads to increased emission of carbon, approximately 98% of that released into the atmosphere, and poses a severe threat to the environment (Balat 2011). The fast depletion of fossil fuel reserves and their impact on the environment have led to the search for alternate sources. In the current situation, biodiesel has received attention as an alternative fuel source. Biodiesel contains esters of fatty acids, which are less toxic than petrochemicals (Kusumo et al. 2017). Biodiesel can be used as a substitute for diesel fuel in existing engines without any modification (Gui et al. 2011; Bhuiya et al. 2016; Pinheiro et al. 2019; Venu et al. 2019). The environmental impact of emissions from biodiesel is less than that of conventional fuels (Sunil et al. 2016; Živković and Veljković 2018). Edible and non-edible oil sources are extensively used for biodiesel production (Sarin et al. 2007; Atabani et al. 2012). *Sterculia foetida* belongs to the Sterculiaceae family. This tree has an average lifespan of 100 years and occurs in tropical and sub-tropical areas (Silitonga et al. 2013). *S. foetida* grows as a tall, straight deciduous tree and attains a height of approximately 40 m and a circumference of approximately 3 m (Vipunngeun and Palanuvej 2009; Carvalho et al. 2012). The fruits of sterculia are very large, and their colour turns from green to red upon ripening and finally to black when they mature. The tree yields approximately 200–350 kg/annually/per tree (Silitonga et al. 2013). *S. foetida* has been found in various parts of the world, such as Bangladesh, Djibouti, Eritrea, Ethiopia, Indonesia, Kenya, Malaysia, Myanmar, Oman, Pakistan, Australia, Philippines, Somalia, Sri Lanka, India, Tanzania, Thailand, Uganda and Republic of Zanzibar (Vipunngeun and Palanuvej 2009). Sterculia seeds have a high oil content of approximately 50–60% by weight (Pandian et al. 2012). The practical methods of oil extraction are steam distillation, cold pressing, mechanical pressing and Soxhlet extraction. Among these methods, Soxhlet extraction is the most suitable method of extracting oil because it yields the maximum amount of oil (Okolie et al. 2012; Oniya et al. 2017). Hence, it is necessary to increase the potential of oil extraction to achieve a high yield of biodiesel. Very recently, n-hexane-based extraction of sterculia oil was found to be efficient compared with the use of other solvents, such as petroleum ether, tetrahydrofuran, methanol and chloroform (Pandian et al. 2012). This implies that there is a gap in the optimisation of oil extraction and biodiesel production, especially from *Sterculia foetida*, which was the primary objective of the present study.

Response surface methodology is a highly useful tool for optimisation studies and for studying the impact of independent parameters and their interactions on the response yield (Mitra et al. 2009; Jamaluddin et al. 2013; Kavitha and Murugavelh 2019b). The usage of these techniques produces an optimum yield (Chen et al. 2005). RSM is used to generate the least number of experiments with optimum conditions compared with the traditional method and to provide a mathematical equation that represents the relationship between the factors and the responses (Giwa et al. 2013; Hashim et al. 2015). In soybean oil extraction, the temperature and flake thickness were the two parameters found to have a large impact on the oil yield (Hashim et al. 2015; Lawson et al. 2010). Currently, artificial neural network (ANN) methods are applied for a wide range of bioprocess optimisation (Manohar and Divakar 2005). The pre-treatment process parameters of biodiesel production were optimised successfully using ANN (Rajendra et al. 2009). The results obtained from the transesterification of yellow oleander oil were evaluated using RSM and ANN. ANN performed better than RSM due to its data fitting and prediction ability (Betiku and Ajala 2014).

Objective of the study

Sterculia foetida is one of the potential sources that can be used as an alternative fuel. There is no significant research on the optimisation of oil extraction from S*terculia foetida.* The objective of the current study is to optimise the factors involved in oil extraction. Response surface methodology was used in this study to determine the significant factors involved in the maximisation of oil yield. The factors involved in the process were the sample weight, volume of seed to solvent ratio and time. The Box-Behnken method was used to find the optimum conditions from the generated experiments. The results obtained from RSM have been validated using ANN. Further, the conventional optimisation of process parameters for biodiesel production and the kinetics of biodiesel production were studied in the current work.

Experimental design

Materials

Sterculia seeds were obtained from the local environment viz. Vellore Institute of Technology, Vellore, India, and soaked in tap water for 4 h to obtain the kernel. The kernel was then dried in a hot air oven, flaked using a blender and sieved using a 1-mm mesh size. Solvent n-hexane (99%), methanol (99.5%) and KOH (85%) pellets were obtained from Merck Laboratories, Mumbai, India. An image of the sterculia kernel is depicted in Fig. 1.

Fig. 1 *Sterculia foetida* kernels after removing outer layer of seed

Response surface methodology

The optimum conditions for oil extracted from sterculia seed kernels were investigated with RSM. The three factors applied for the optimisation of extraction were the sample amount (g), seed to solvent ratio (g/mL) and extraction time (h), as shown in Table 1. Design expert 10.0.3 software reported 17 runs for optimising the factors in this process, as shown in Table 2. The Box-Behnken design was employed to optimise the oil extraction from sterculia seed kernels.

Oil extraction from Sterculia foetida by using N-hexane

Oil extraction was performed based on the polarity of the solvent and property of the seeds. The efficiency of the extraction process depends on the selection of the solvent. Oil yield was calculated using the formula given below.

$$\text{Oil yield} = \frac{\text{Mass of extracted oil}}{\text{Mass of sample}} \times 100 \quad (1)$$

FFA analysis

One gram of oil sample was taken in a conical flask and titrated against 0.1 N methanolic KOH. One hundred millilitres of neutral alcohol was mixed with the oil, and a few drops of phenolphthalein were added as an indicator. The oil sample was titrated until a light pink colour appeared. The FFA content can be calculated from the formula given below.

$$\text{FFA\%} = \frac{(\text{Titrant mL}) \times (\text{N of titrant}) \times (\text{Avg.Mwt.of fatty acid})}{(\text{Sample weight}) \times (10)} \quad (2)$$

Table 1 Range of process variables

Factors	Symbol	Coded variables		
		− 1	0	+ 1
Sample weight (g)	A	5.0	7.5	10
Solvent to seed ratio (ml/g)	B	20	40	60
Extraction time (h)	C	2	3	4

Statistical data analysis

Statistical analysis of the experimental data was performed by RSM by fitting a suitable quadratic polynomial equation derived from Design Expert 10.0.3. The output variable was related to the coded factors by using multiple regressions to evaluate the polynomial model of the output. The model was evaluated using a significance test and analysis of variance (ANOVA).

Utilisation of ANN

MATLAB R2016a software provides a neural network tool that was used for the present study. A multilayer artificial neural network comprised an input layer, a hidden layer and an output layer. The ANN architecture had an input layer with three neurons, an output layer with one neuron and a hidden layer with 30 neurons. The input and output data range were from − 1 to + 1, which was considered by the neural network. All the layers were processed through the neural network from the given input data (sample amount, seed to solvent ratio and extraction time) to the output data, the extracted oil yield (Y). The output was predicted using a feed-forward backpropagation neural network, which was trained using the Levenberg-Marquardt algorithm. Tansig and the gradient descent method were utilised as the transfer function and adaption learning function in this study.

Sterculia fatty acid analysis

The fatty acid composition of *Sterculia foetida* was analysed by converting the oil into sterculia fatty acid methyl esters using a methanolic solution (2% sulphuric acid in methanol). This mix was refluxed for 4 h at 70 °C to obtain fatty acid methyl ester, which was further processed by treating it with ethyl acetate. The final dried product was subjected to GC-MS analysis.

An Agilent 6890 gas chromatograph was used to detect the composition of fatty acids in the oil with a non-bonded cyano silicone column. The oven temperature was set to 160 °C for 2 min to stabilise the instrument and was gradually increased from 180 to 230 °C. The sample was injected when the temperature reached 250 °C.

Table 2 Experimental runs with observed yield and predicted yield of oil

RUN	Sample weight (g)	Solvent to seed ratio (ml/g)	Extraction time (h)	Observed yield (w/w%)	Predicted yield (w/w%) RSM	Predicted yield (w/w%) ANN
1	10	20	3	23.96	23.61	20.92
2	7.5	60	2	39.96	39.78	39.63
3	7.5	40	3	45.08	45.27	45.72
4	7.5	40	3	45.18	45.27	45.72
5	5	40	2	19.54	19.37	20.13
6	7.5	20	2	14.85	14.74	15.31
7	5	40	4	39.4	38.94	39.34
8	10	60	3	34.8	34.52	34.75
9	7.5	40	3	44.8	45.27	45.72
10	10	40	2	35.44	35.91	34.69
11	10	40	4	20.12	20.29	19.93
12	7.5	40	3	45.12	45.27	45.72
13	7.5	60	4	24.44	24.55	24.26
14	7.5	20	4	33.74	33.92	34.78
15	5	60	3	32.14	32.49	45.62
16	5	20	3	27.45	27.73	27.27
17	7.5	40	3	46.15	45.27	45.72

Degumming of oil

Extracted sterculia oil was degummed by adding 5% hexane to the oil, which was then heated to 40 °C and stirred for 20 min (Devan and Mahalakshmi 2009a). This process removes all the unwanted impurities, gums, which settle at the bottom layer. The top layer consists of purified oil for biodiesel production.

Transesterification process

Degummed sterculia oil with an FFA content of 3% was converted to biodiesel using single alkali transesterification. For this purpose, 884 g of sterculia oil and 1.5% KOH (weight basis) catalyst in methanol (1:6 mol/mol ratio) were placed in a 2000-mL round bottom flask and heated to 55 °C for an hour using an agitator at a speed of 600 rpm. Finally, the sterculia biodiesel was subjected to separation. The upper (biodiesel) and lower (glycerol) layers were separated. Warm water was added to purify the biodiesel. The washed biodiesel was heated to 100 °C and stored for further analysis.

Physicochemical properties of oil and biodiesel

Specific characteristics of sterculia oil and biodiesel from *Sterculia foetida* were analysed. The parameters tested were the flash point, viscosity, fire point, calorific value, pour point, acid value, saponification value, cloud point and density, as shown in Table 3.

Optimisation of biodiesel production

The optimum conditions for maximum biodiesel production were analysed. Temperature, feedstock to methanol ratio, catalyst percentage (KOH) and conversion time are the selected parameters that were varied to define the optimum conditions for the production of biodiesel. The transesterification temperature ranged from 45 to 65 °C, the oil to methanol ratio ranged from 1:3 to 1:15, the catalyst percentage ranged from 0.75 to 2% and the transesterification reaction time ranged from 30 to 120 min. The impact on the yield of biodiesel was studied.

Kinetics of biodiesel production

In our current study, the kinetics of *Sterculia foetida* methyl ester (SFME) production were investigated. This kinetic study helped to determine the rate constant and feasibility of chemical reactions. The excess methanol concentration leads to the formation of methyl esters, and thus, the reaction exhibits a pseudo-first-order mechanism (Bisheswar et al. 2018). The rate constant was obtained by plotting the product concentration against time. Finally, the kinetic equation was expressed as:

$$\frac{d[B]}{d[t]} = K[B] \tag{3}$$

where B represents the yield percentage of SFME (%), t represents the reaction time for biodiesel production in min and k represents the rate constant of this reaction (min^{-1}). The plot of ln $[B]$ and ln $[dB]/[dt]$ was observed to be linear, and

Table 3 Physicochemical properties of oil and biodiesel

S. no	Physicochemical property	Sterculia oil	Sterculia biodiesel	Munurung et al. 2012	Devan and Mahalakshmi 2009b	Silitonga et al. 2013	ASTM standards	ASTM 6751
1.	Density (kg/m^3)	955 ± 0.1	870 ± 0.1	873	875	879.1		880
2.	Kinematic viscosity at 40 °C in (mm^2/s)	42 ± 0.006	5.1 ± 0.006	4.92	6.0	3.96	ASTM D445	1.9–6.0
3.	Acid value (mg KOH/g)	6.171 ± 0.004	0.34 ± 0.004	–	–	0.14	ASTM D664	0.5
4.	Acidity	0.11	0.03	–	–		ASTM D664	–
5.	Free fatty acid (%)	3.10	0.8	–	–		ASTM D664	–
6.	Cloud point (°C)	1 ± 1	− 3 ± 1	–	–	− 3	ASTM D2500	− 3 to − 12
7.	Pour point (°C)	− 3 ± 1	− 3 ± 1	− 3	1	− 3	ASTM D2500	− 15 to − 16
8.	Calorific value (MJ/kg)	36.446	38.390	40.167	40.211		ASTM D-1826	–
9.	Oil yield (%)	45%	–	–	–	–	–	–
10.	Flash point (°C)	–	162 ± 0.5	160.5	162	160	ASTM D92-16b	100–170
11.	Fire point (°C)	–	178 ± 0.5	–	–		ASTM D92-16b	–

the slope determines the rate constant of the reaction. The reaction rate was found to increase with increasing temperature (Endalew et al. 2011).

The equation obtained by integrating both sides is given below:

$$\ln[A]_t - \ln[A]_0 = -k(t-0) \tag{4}$$

Finally, arranged in the form y = mx + C:

$$\ln[A]_t = -kt + \ln[A]_0 \tag{5}$$

Time and temperature were the parameters that increased the reaction rate. The activation energy required for this reaction can be calculated using the Arrhenius equation. The connection between the reaction rate constant and the temperature is expressed as:

$$K = Ae^{-\frac{Ea}{RT}} \tag{6}$$

Here, k indicates the reaction rate constant (min^{-1}), A indicates the Arrhenius constant, E_a represents the activation energy (J/mol), R is the universal gas constant and T is the absolute temperature (K). The plot of ln (k) and $1/T$ gives the (slope) activation energy ($-E_a/R$), and the intercept gives the Arrhenius constant, ln A (Suganya et al. 2013). The activation thermodynamic parameters were estimated according to transition state theory using the equations below:

$$A = \frac{RT}{Nh} e^{\frac{\Delta S^{++}}{R}} \tag{7}$$

N is Avogadro's constant, h is Planck's constant, ΔS^{++} is the activation entropy, ΔH^{++} is the activation enthalpy and ΔG^{++} is the Gibbs free energy or activation energy.

$$\Delta H^{++} = E_a - RT \tag{8}$$

$$\Delta G^{++} = \Delta H^{++} - \Delta S^{++} \tag{9}$$

$$Y = \frac{B_t}{B_u} \tag{10}$$

$$\ln Y = -\frac{\Delta G}{R}\frac{1}{T} = -\frac{\Delta H}{R}\frac{1}{T} + \frac{\Delta S}{R} \tag{11}$$

Y is the equilibrium constant, B_t is the biodiesel yield with respect to temperature, B_u is the oil not converted into biodiesel, ΔS is the entropy change, ΔH is the enthalpy change and ΔG is Gibb's free energy. In the plot of ln B_t versus 1/T, concerning to the time and the slope obtained from the straight line indicate the enthalpy change ΔH for biodiesel production.

Results and discussion

Fatty acid percentage in sterculia oil

The methyl ester composition by percentage varies with the feedstock quality, geological location and growth conditions of the species (Atabani et al. 2014). The percentages of fatty acids in sterculia seed oil are reported in Table 4. The methyl ester composition of *Sterculia foetida* oil included approximately 32% sterculic acid. The percentages of palmitic acid, oleic acid and linoleic acid were reported to be approximately 15.88%, 10% and 9.95%, respectively, as shown in Fig. 2a. The percentage of malvalic acid present in the oil was approximately 9%. The percentage of eicosenoic acid was found to be 4.81%, and arachidic acid was 4.07%. Eicosenoic acid was reported to be present in *Sterculia urens* at 1.99% (Galla et al. 2012). The percentage of stearic acid was found to be 3.30%. The

Table 4 Fatty acid composition of *Sterculia foetida* seed oil

Fatty acid	Structure	Systematic name	Wt%
Lauric acid	$C_{12:0}$	Dodecanoic acid	0.97%
Myristic acid	$C_{14:0}$	Tetradecanoic	0.53%
Palmitic acid	$C_{16:0}$	Hexadecanoic	15.88%
Stearic acid	$C_{18:0}$	Octadecanoic	3.30%
Oleic acid	$C_{18:1}$	*cis*-9-octadecenoic acid	10.00%
Linoleic acid	$C_{18:2}$	Cis-9-cis-12 octadecadienoic	9.95%
Sterculic acid	$C_{19:3}$	8-(2-Octylcyclopropen-1-yl)octanoic acid	32%
Malvalic acid	$C_{18:3}$	7-(2-Octylcyclopropen-1-yl)heptanoic acid	9%
Arachidic acid	$C_{20:0}$	Eicosanoic	4.07%
Eicosenoic acid	$C_{20:1}$	*cis*-11-eicosenoic acid	4.81%
Behenic acid	$C_{22:0}$	Docosanoic acid	0.51%
Unidentified fatty acid	–	–	8.98%
Degree of unsaturation			152.9
Long chain saturated fatty acid			16.49
Saturated fatty acid			25.26
Monounsaturated fatty acid			55.81
Polyunsaturated fatty acid			9.95

other fatty acids were lauric acid, myristic acid and behenic acid. The percentage of palmitic acid obtained matches a reported value of 10.5 to 17.7% (Silitonga et al. 2013). Degree of unsaturation and long chain saturated factor was calculated and given in the Table 4. Table 5 shows the comparison of the methyl ester composition of *Sterculia foetida* with the values obtained by other authors.

In this section, the optimisation of oil extraction is reported. The predicted and observed yields of the model are depicted in Table 2. The data obtained from the experimental results were statistically analysed using analysis of variance. The ANOVA results of oil extraction yield are shown in Table 6. The effects of sample amount, seed to solvent volume ratio and extraction time were found to be significant, since the P values were less than 0.05. The results revealed that the P values of the model were significant. The F value was 7.54, which indicates that the model was significant. The strongest influencing factor in the model was the solvent to seed ratio. The standard deviation, mean and coefficient of variance (c.v.) were estimated to be 0.54, 33.66 and 1.61, respectively, as shown in Table 7. R^2 was 0.9988, adj. R^2 was 0.9972 and an adequate precision of 73.449 was obtained for the oil extraction yield. The adjusted R^2 value was 0.9972, which supports the significance of the model. The lack of fit was non-significant, which implies that the model fits well. Hence, this model can be used in the analysis of oil yield. The equation for the actual variables in the Box-Behnken model is given in Eq. (11). Here, A represents sample weight (g), B represents the seed to solvent ratio and C represents extraction time.

$$\text{Oil yield} = -267.38 + 27.47 \times A + 2.86 \times B + 98.52 \times C + 0.0307 \times A \times B - 3.51 \times A \times C - 0.43 \times B \times C - 1.22 \times A^2 - 0.02 \times B^2 - 8.99 \times C^2 \quad (12)$$

Impact of the sample amount

The ratio of seed to solvent volume was the most influential parameter related to the sample amount. The optimum sample amount was found to be 7.5 g, as shown in Fig. 2b. Increasing the sample amount decreased the amount of oil yield. This might be because the amount of solvent vapour exposed to the sample was lower when a large amount of sample was taken. The lower concentration of solvent led to a lower oil yield, whereas a higher concentration of solvent increased the oil yield (Oniya et al. 2017). A sample amount of 7.5 g and a 40 mL/g ratio for 3 h resulted in 45.08% biodiesel, whereas changing the concentration of solvent to 20 mL/g yielded 33.74% with the same sample amount. This result can be attributed to the fact that the amount of solvent vapour was less than needed to interact with the active sites of the sample. Variation in the solvent concentration has a high impact on the oil yield. The combined effect of sample amount (g) and the seed to solvent volume ratio (g/mL) had a significant effect on the oil yield from sterculia seed kernels.

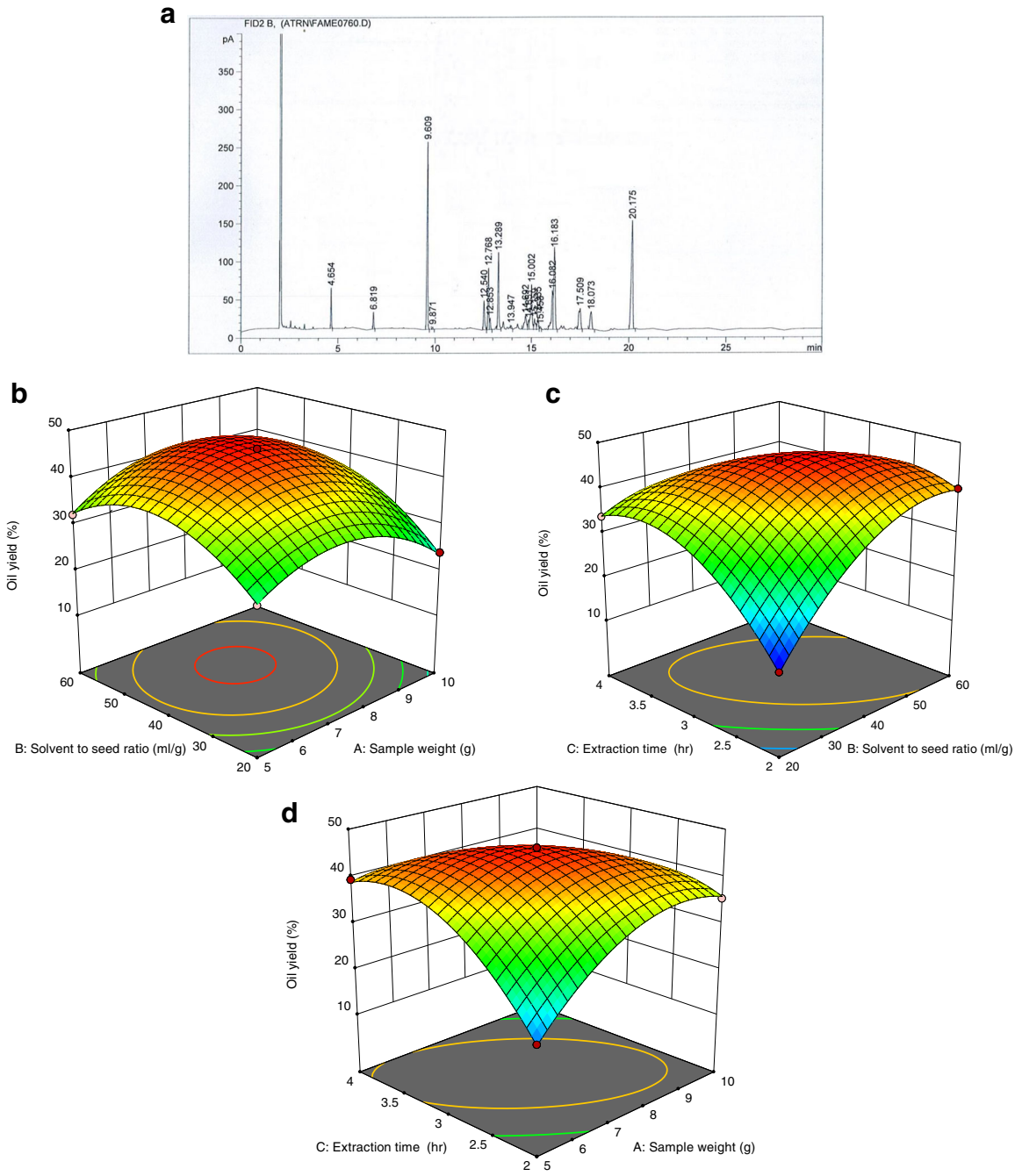

Fig. 2 **a** GC-MS chromatograph of sterculia oil. **b** 3D surface plot showing the effect of sample weight and seed to solvent ratio. **c** 3D surface plot showing the effect of seed to solvent ratio and extraction time. **d** 3D surface plot showing the effect of sample weight and extraction time

Impact of the extraction time

The impacts of the sample amount and extraction time are shown in Fig. 2c. An increase in the sample amount decreased the oil yield, even though the extraction time was increased to 4 h (Oniya et al. 2017). This might be due to the competition between the sample and the solvent. The solvent utilises the active pores in the sample by the proximity effect. A decrease in yield was found to occur with an increase in the sample amount. This may be due to the increase in the number of unutilised pores in the sample. The optimum extraction time was found to be 3 h, and the sample amount was 7.5 g. Thus, the sample amount has a significant impact on the oil yield.

Impact of the seed to solvent volume ratio

The impacts of the seed to solvent volume ratio and extraction time are shown in Fig. 2d. The maximum solvent to sterculia

Table 5 Comparative study on fatty acid composition of *sterculia foetida*

S. no	Fatty acid name	Percentage of fatty acid	References
1	Oleic acid	6.0	(Silitonga et al. 2013)
2	Linolenic acid	0.2	(Sabria et al. 2004)
3	Palmitic acid	15.0	(Sabria et al. 2004)
4	Sterculic acid	45.3	(Kale et al. 2011)
5	Stearic acid	4.7	(Silitonga et al. 2013)
6	Linoleic acid	9.1	(Silitonga et al. 2013)
7	Myristic acid	0.2	(Silitonga et al. 2013)
8	Lauric acid	0.97	(Silitonga et al. 2013)
9	Arachidic acid	2.3	(Silitonga et al. 2013)
10	Malvalic acid	5.4	(Sabria et al. 2004)

seed ratio was found to be 40 mL/g. The amount of oil extracted from 7.5 g of seed with 20 mL/g of hexane solvent was a 14.85% yield. A further increase in the solvent volume to 40 mL/g increased the yield to 44.8%. The maximum oil yield was observed when the solvent volume was increased, which was due to the full utilisation of active sites in the sample by the hexane. A further increase in the solvent volume to 60 mL/g for 7.5 g of sample resulted in an oil yield of 24.4%. A drop in the yield was noted with an increase in the volume of solvent to 60 mL, which might be because all available pores utilised by the solvent. The excess solvent used did not take part in the reaction. Thus, the optimum concentration of the solvent avoids the use of excess solvent. A similar trend was observed for all the sample amounts studied in Table 2. Biodiesel was produced using the purified degummed oil. After purification, the biodiesel was heated to 100 °C and stored for analysis. The experiments were performed three times, and the yield obtained was approximately ± 90.19%.

Validation of the model

RSM has been adopted to optimise the parameters and to obtain the response. The optimised values, which correspond to the highest desirability of 0.989, were chosen and confirmed by performing experiments. The details of the experiments conducted using these optimised values are given below in Table 8. Values obtained for the same set of experiments were near the predicted values, and thus, the obtained optimum values of the factors and yield were confirmed. Thus, the model has been validated. The maximum yield obtained was 45.814%.

Testing and validation using ANN

The Levenberg-Marquardt algorithm has numerous advantages over other algorithms because of its high R^2 value and low deviation and MSE (mean square error). Model validation in ANN was performed based on the R^2 value of the experimental yield and predicted yield. The R^2 values for the training, validation and testing sets and all data were 0.9992, 0.9999, 0.8218 and 0.9508, respectively. The ANN predicted values and experimental values were found to be compatible. In the present study, data obtained from experiments were randomly divided into training, validation and testing sets. The training set comprises 70% of the data, the validation set contains 15% of the data and the testing set contains 15% of the data. The variation in the mean square error value for the developed ANN model, as shown in Fig. 3a. The experimental values and the predicted values of oil yield found by the ANN model are shown in Fig. 3b.

Table 6 Analysis of variance for model regression

Source	Sum of Squares	Df	Mean square	*F* value	*P* value
Model	1706.11	9	189.57	645.53	< 0.0001
A-Sample amount	2.22	1	2.22	7.54	0.0286
B-Solvent to seed ratio	122.77	1	122.77	418.08	< 0.0001
C-Extraction time	7.82	1	7.82	26.63	0.0013
AB	9.46	1	9.46	32.20	0.0008
AC	309.41	1	309.41	1053.61	< 0.0001
BC	296.01	1	296.01	1008.00	< 0.0001
A^2	246.44	1	246.44	839.20	< 0.0001
B^2	271.36	1	271.36	924.06	< 0.0001
C^2	340.33	1	340.33	1158.92	< 0.0001
Residual	2.06	7	0.29		
Lack of fit	0.99	3	0.33	1.25	0.4035
Pure error	1.06	4	0.27		
Cor total	1708.16	16			

Table 7 Response of the model

Std. dev.	0.54	R^2	0.9988
Mean	33.66	Adj R^2	0.9972
C.V. %	1.61	Pred R^2	0.9897
PRESS	17.56	Adeq precision	73.449

Table 8 Validation of experiments

S. no	Sample amount (g)	Seed to solvent ratio (ml/g)	Extraction time (h)	Predicted value	Response 1	Response 2	Response 3
1.	7.69	46.512	2.86	46.00	45.69	45.71	45.84

Impact of temperature

The transesterification reaction temperature plays a significant role in biodiesel yield. An increase in temperature elevates the reaction rate, and the solubility of catalyst in methanol was also achieved with increasing temperature (Muthukumaran et al. 2017). The temperature was varied from 45 to 65 °C (in 5 °C intervals). At 55 °C, the yield of biodiesel was found to be higher than at other temperatures. The yield of biodiesel was found to be directly proportional to the increase in

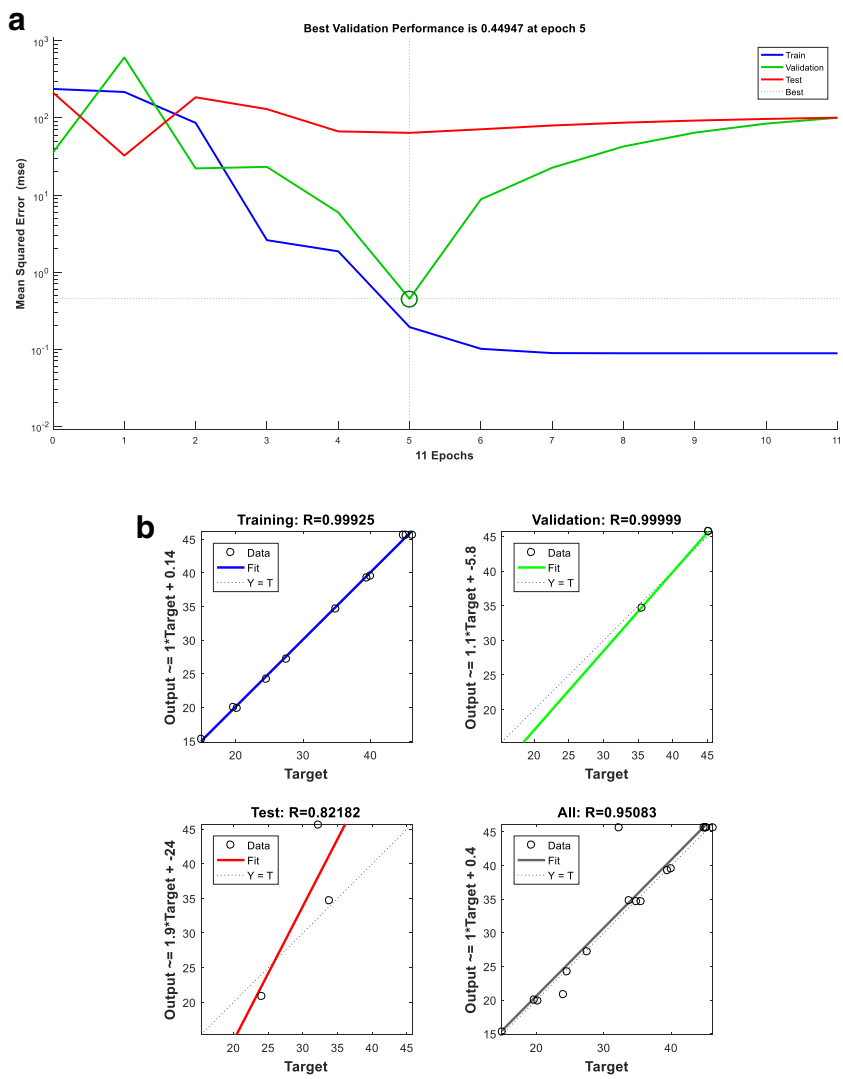

Fig. 3 **a** MSE variation. **b** Experimental and predicted values of ANN

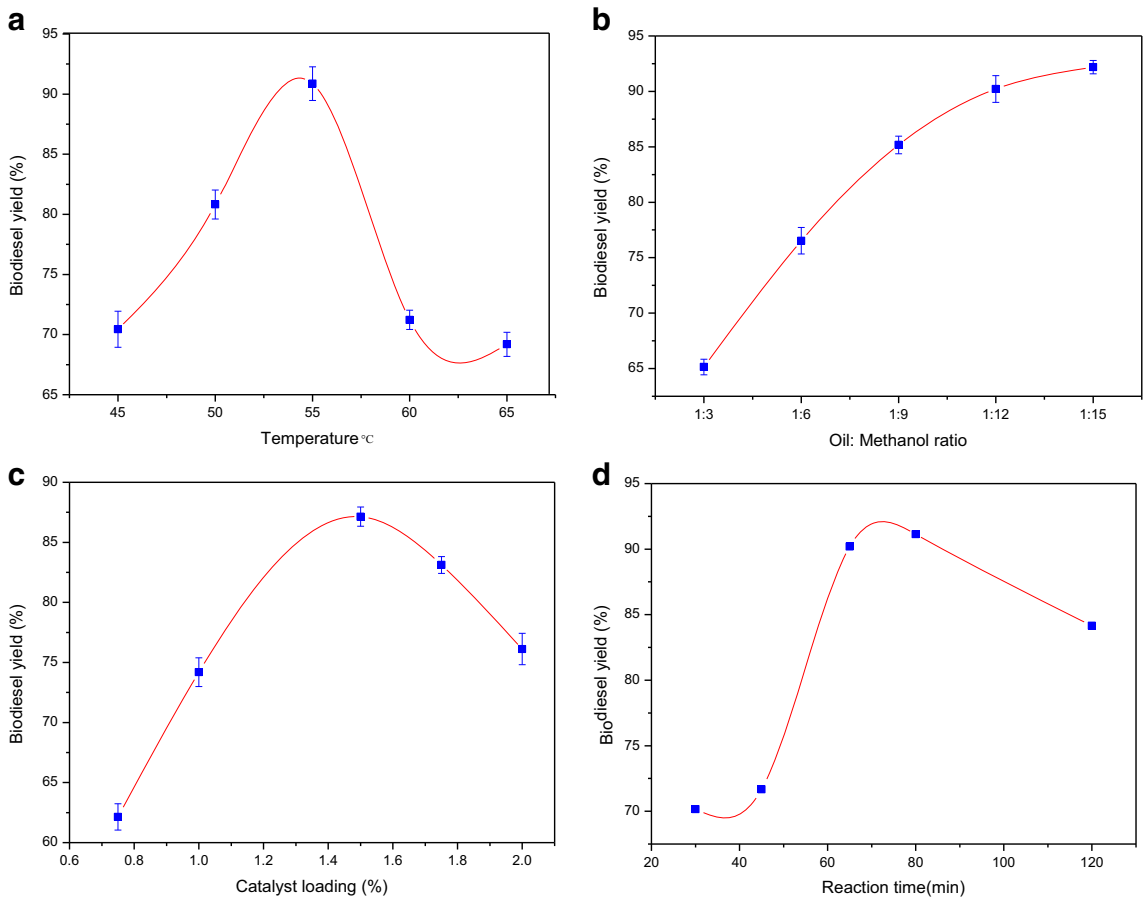

Fig. 4 **a** Effect of temperature on biodiesel yield. **b** Effect of oil to methanol ratio on biodiesel yield. **c** Effect of catalyst loading on biodiesel yield. **d** Effect of reaction time on biodiesel yield

temperature. Figure 4a shows the relationship between temperature and biodiesel yield: 90.87(V/V) % biodiesel was acquired at 55 °C, which was the highest yield.

Impact of oil to methanol ratio

An appropriate methanol concentration for the transesterification process results in better yield. The methanol concentration is an important parameter because it determines the compatibility of the reaction and the selected solvent. The solubility of free fatty acids in the solvent favours the formation of biodiesel. In homogeneous reaction, the catalyst solubility also depends on the volume of methanol concentration (Muthukumaran et al. 2017). Figure 4b represents the yield of biodiesel versus feedstock to methanol ratio. This optimum feedstock to methanol ratio required for transesterification was found to be 1:15, which yields SFME of approximately 92.19%. At a 1:12 ratio, the yield was found to be 90.22%. The yield was found to be lower, at a ratio of 1:3, at approximately 65.14%, which might be due to an insufficient amount of methanol for the reaction. The increase in yield at 1:15 was due to the interaction of the methanol vapour with the free fatty acid to form esters. The yield of biodiesel was found to increase by 2% when the oil to methanol ratio was increased from 1:12 to 1:15.

Table 9 Biodiesel yield with respect to the temperature and reaction time

Time (min)							
Temperature (K)	30	45	65	80	95	*k* value	R^2
318	76.39	81.52	84.86	87.02	91.85	2.7×10^{-3}	0.9752
323	76.59	80.82	85.88	88.89	93.40	3.2×10^{-3}	0.9960
328	72.47	77.82	84.88	89.29	93.75	4.0×10^{-3}	0.9863
333	69.04	74.12	81.84	85.62	94.32	5.1×10^{-3}	0.9940

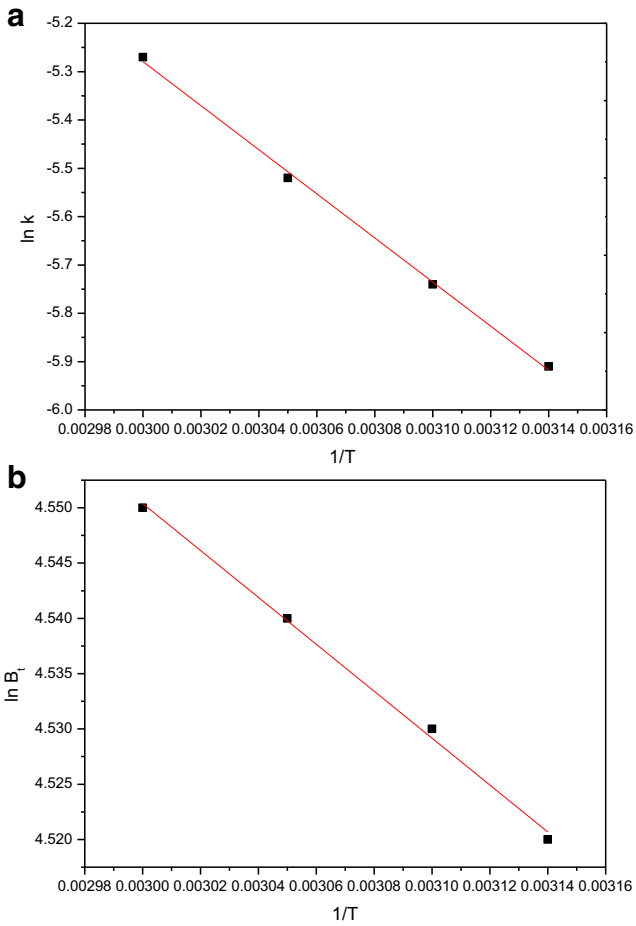

Fig. 5 **a** Graph between ln k verses 1/T. **b** Graph between ln B_t verses $1/T$

Impact of catalyst loading

The catalyst percentage (KOH) was tested at values of 0.75, 1.0, 1.5, 1.75 and 2%, as shown in Fig. 4c. The maximum amount of ester was obtained at 1.5% catalyst loading. The higher concentration of 2% catalyst loading resulted in a decreased biodiesel yield. At 1.5% catalyst loading, an 87.14% yield of sterculia biodiesel was obtained. When 1.75% catalyst was utilised, the methyl ester formed was 83.12%, and when the catalyst loading was 2%, the biodiesel yield was 76.12%. A higher concentration of catalyst loading resulted in a lower amount of product. This behaviour may be due to an increase in viscosity, or it may lead to a backward reaction, thus producing glycerol and soap (Kumar et al. 2018). The higher catalyst concentration leads to precipitation in the reaction mixture.

Impact of reaction time

The time was varied from 30 to 120 min to study the effect on biodiesel yield, as shown in Fig. 4d. The optimum time required for the production process was found to be 65 min, which yields approximately 90.19% biodiesel. An increase in the reaction time to 80 min increased the yield to 91.13%, whereas the yield decreased to 84.16% at 120 min. This result implied that biodiesel production is a reversible reaction, and at the equilibrium stage, there was a decline in the biodiesel yield. Thus, the optimum reaction time was found to be 65 min.

Property of oil and biodiesel

The properties of sterculia oil and sterculia biodiesel were analysed. The results are shown in Table 3. The density of the oil was 955 kg/m^3, the viscosity was 42 mm^2/s, the cloud point 1 °C, the pour point − 3 °C and the calorific value 38.390 MJ/kg. The results obtained were compatible with previous findings (Atabani et al. 2013). The flash point and fire point of the oil were 296.6 °C and 313.4 °C, respectively. Viscosity refers to the thickness of the oil, which has been analysed for better engine performance. High viscosity forms deposits on the engine and has a significant impact on the fuel injection system (Atabani et al. 2013; Atabani and da Silva 2014). The flash point of biodiesel is in a higher range than that of diesel, since it helps in safe transportation, and the flash point obtained for biodiesel from *Sterculia foetida* is in line with values found by other researchers (Silitonga et al. 2013). The appearance of cloud crystals in the fuel at a specific temperature is referred to as the cloud point, and the formation of gel at a specified temperature is known as the pour point (Atabani et al. 2012). These values were also close to those obtained by other researchers. Calorific values were also found to be an important parameter, and as per the standards, the minimum calorific value of the fuel should be 35 MJ/kg.

Table 10 Thermodynamic parameters and equilibrium constants at various temperatures

S. no	T (K)	ΔH^{++} (KJ mol^{-1})	ΔS^{++} (Jmol^{-1} K^{-1})	ΔG^{++} (KJ mol^{-1})	Y	ΔS (mol^{-1} K^{-1})	ΔG (KJ mol^{-1})
1	318	35.26	− 239.19	76.09	11.27	20.11	− 5.92
2	323	35.22	− 239.32	77.33	14.15	22.02	− 7.11
3	328	35.18	− 239.45	78.57	15	22.52	− 7.39
4	333	35.14	− 239.58	79.81	16.60	23.35	− 7.77

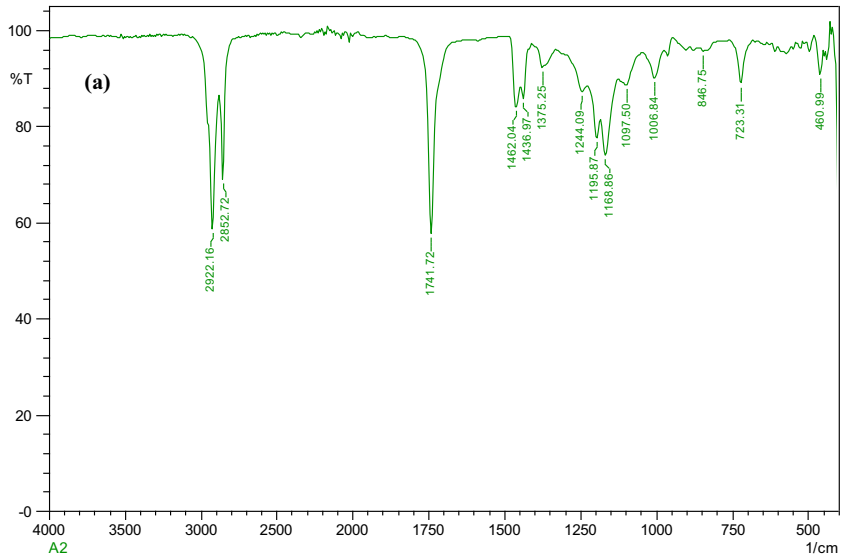

Fig. 6 FT-IR analysis of Sterculia biodiesel

Kinetics of SFME

The kinetics of biodiesel production from sterculia oil were investigated with different temperatures and reaction times. Temperatures of 40, 50, 55 and 60 °C were tested in the transesterification process, and the reaction time was also varied to study the kinetics of biodiesel production. The optimum condition of the process was studied, and the data of biodiesel production fit the first-order kinetics. The yield of biodiesel for a different temperature, reaction time and rate constant value, and R^2 value is given in Table 9.

A graph of ln k versus $1/T$, produced a straight line, as shown in Fig. 5a. The slope obtained from this line gives the activation energy. The activation energy required for biodiesel production of sterculia oil using KOH as a catalyst was calculated to be 37.91 kJ mol^{-1}. The enthalpy change for biodiesel production from *Sterculia foetida* for 95 min was obtained by plotting the graph of Y_T versus $1/T$, which is shown in Fig. 5b. The slope of this graph gives the enthalpy change for biodiesel production, which was found to be 1.76 kJ mol^{-1}. The enthalpy values were found to be positive, which indicates that the reaction was endothermic and required energy for the process, as shown in Table 10.

FT-IR analysis

The FT-IR analysis of biodiesel from *Sterculia foetida* is shown in Fig. 6. The strong triplet bands in the region of 2980–2800 cm^{-1} indicate the presence of methyl and methylene lipids. A strong ester band (C=O) was observed in the region 1710 cm^{-1}, which specifies the conversion of oil into biodiesel. The spectral bands at 1710 cm^{-1} and 2800 cm^{-1} confirmed biodiesel conversion (Rashid et al. 2011). The peak at 720 cm^{-1} indicates the presence of long chain fatty acids, as observed in rich chia seed oil (Timilsena et al. 2017).

Conclusion

The primary fatty acids reported in *Sterculia foetida* seed were sterculic acid (32%), palmitic acid (15.88%), oleic acid (10.00%), linoleic acid (9.95%) and malvalic acid (9%). The minor fatty acids observed in the oil sample were lauric acid (0.97%), myristic acid (0.53%) and behenic acid (0.51%). Response surface methodology was utilised to optimise oil extraction from *Sterculia foetida* seeds. The Box-Behnken method predicted the optimum values of the factors to produce a high oil yield. The seed to the solvent ratio was an important parameter with the maximum impact on the oil yield. Decreasing the volume of solvent minimises the oil yield, whereas using a high amount of the solvent increases the cost of the process. The extraction time was also optimised and minimised. The oil yield was also maximised, which was another important goal of this experiment. The most favourable seed to the solvent ratio was found to be 1:4. The ideal sample amount was also predicted to be approximately 7.5 g. The maximum yield was obtained at 3 h. The obtained results were found to be valid since the values of the experimental and predicted values were almost the same. The maximum oil yield was 46.15%. The physicochemical characteristics of oil were also analysed, and the results were tabulated. The maximum yield from biodiesel optimisation was obtained at a temperature of 55 °C, molar ratio of 1:12, catalyst loading of 1.5% and reaction time of 80 min. Biodiesel production from *Sterculia foetida* using KOH as the catalyst requires an activation energy of 37.91 kJ mol^{-1}. The enthalpy value was found to be positive, which represents an endothermic

reaction, and energy is required for the biodiesel production process.

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

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